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Modern Datacenter Architecture for Pharmaceutical Companies: Scalability, Security, and Compliance

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THE MODERN DATACENTER IN THE PHARMACEUTICAL INDUSTRY



datacenter infrastructure pharmaceutical cloud-computing hybrid-cloud security
compliance gxp data-management high-performance-computing edge-computing
digital-transformation it-strategy regulatory-compliance

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Pharmaceutical Industry

Introduction

In the pharmaceutical sector, data is as critical as any chemical reagent. Modern drug discovery, clinical development, and manufacturing all depend on robust computing infrastructure. The **modern datacenter** in pharma goes far beyond a room of servers – it's an ecosystem encompassing on-premises facilities, cloud services, and advanced software platforms. Today's pharma datacenters must handle massive scientific datasets, enable rapid analysis (often in real-time), and meet rigorous regulatory requirements. This article explores how modern datacenters have evolved in pharma, what they comprise, and how they support cutting-edge workloads while ensuring compliance, security, and scalability. We'll also compare cloud, on-prem, and hybrid architectures, discuss sustainability trends, review key vendors/tools, and outline best practices for datacenter modernization in the pharma industry.

Evolution of the Modern Datacenter in Pharma

Pharmaceutical companies have long been technology-intensive, but their datacenters have transformed dramatically over the past two decades. **Early 2000s:** Pharma IT infrastructure was dominated by on-premises servers and specialized high-performance computing (HPC) clusters for R&D. These traditional datacenters were hardware-centric, with siloed storage and compute, often running on proprietary systems or dedicated machines for each application. Over time, **virtualization** emerged – hypervisors like VMware allowed consolidating many virtual servers on one physical host, improving utilization and agility. Virtualization became mainstream in pharma by the 2010s, enabling more efficient use of hardware and easier provisioning of environments for labs and enterprise apps.

The **cloud computing era** then arrived, albeit cautiously for pharma. Initial concerns about data security and regulatory compliance made pharma companies slow to embrace cloud. However, the value proposition – on-demand scalability and reduced IT overhead – became hard to ignore. By 2025-2026, cloud adoption in pharma has become nearly universal: *about 83% of pharmaceutical companies leverage cloud solutions in some form*, with 40% of pharma and life sciences firms



valued at \$18.3 billion in 2024 and is projected to reach \$62.39 billion by 2033, growing at a CAGR of 14.6% ⁽¹²⁾. Cloud technology played a key role in accelerating the development of critical therapies; for example, leveraging cloud platforms helped deliver the first COVID-19 vaccine candidate to clinical trials in under 50 days ⁽¹¹⁾, an unprecedented timeline. This demonstrated how a modern cloud-enabled datacenter can vastly speed up innovation.

Today's **modern datacenter** in pharma is often a **hybrid** one – blending on-premises systems (still crucial for certain sensitive or high-performance workloads) with public cloud services. Data and workloads flow between on-site clusters and cloud instances, giving pharma firms the flexibility to burst to the cloud for peak demands or specialized services (like AI platforms) while keeping core systems under direct control. The datacenter itself has also "software-defined" components: software-defined storage and networking, containerization, and orchestration have become common, making infrastructure more agile and easier to manage via code. Another recent evolution is the shift to **hyper-converged infrastructure (HCI)**, which combines compute and storage into modular units. In one case study, a midsize pharma company migrated legacy systems to an HCI platform and reduced their data center footprint to 40% of its original size, cutting power consumption by 50% and slashing management effort by 80% ⁽¹³⁾ ⁽³¹⁾. This kind of modernization illustrates the efficiency gains now achievable. Overall, the pharma datacenter has evolved from a static, hardware-bound environment to a dynamic, hybrid cloud-enabled platform that is integral to digital transformation in life sciences.

Key Infrastructure Components of a Modern Pharma Datacenter

Modern datacenters are composed of several key infrastructure layers and components, each of which has advanced to meet the demands of pharmaceutical computing:

Compute (Servers and HPC Nodes)

Compute power is the foundation – this includes everything from general-purpose servers to specialized HPC nodes and accelerators. Pharma datacenters still deploy racks of **x86 servers** (often with multi-core Intel Xeon or AMD EPYC processors) for enterprise applications (e.g. ERP, LIMS, clinical data systems) and basic database/workflow processing. However, many R&D and analytics workloads demand HPC capabilities. This has led to clusters of compute nodes with high-



Processing Units) and other accelerators are deployed alongside CPUs to speed up AI and simulation workloads – a single NVIDIA GPU can perform certain drug discovery computations orders of magnitude faster than a CPU core. Modern pharma datacenters often integrate GPU-based servers for AI-driven research (for example, training a machine learning model to analyze medical images or predict protein folding). NVIDIA offers domain-specific platforms like **NVIDIA Clara** and the expanded **NVIDIA BioNeMo** platform for healthcare and life sciences. In January 2026, NVIDIA announced a major expansion of BioNeMo at the J.P. Morgan Healthcare Conference, introducing new Clara open models including RNAPro for RNA structure prediction and ReaSyn v2 for ensuring AI-designed drugs are practical to synthesize ([4]). A landmark collaboration was announced between NVIDIA and Eli Lilly – a \$1 billion co-innovation AI lab based in San Francisco, representing one of the largest disclosed AI partnerships in pharma ([5]). The lab will be built on the NVIDIA BioNeMo platform and the new NVIDIA Vera Rubin architecture. This highlights how essential accelerated computing has become in pharma IT, with companies like Bristol-Myers Squibb reporting 55% overall cost savings from deploying NVIDIA DGX SuperPOD systems for oncology R&D.

To maximize compute utilization, datacenters rely on **virtualization** and **containers** (discussed further below). It's common for dozens of virtual machines or containers to run on a single physical host, isolating different applications or tenants while sharing the underlying physical CPUs/RAM. Pharma research groups might use a virtualized HPC cluster where each team gets virtual nodes carved out of the physical cluster, or use containerized workloads that can be scheduled onto any available server. The net effect is far better hardware utilization and flexibility than the dedicated servers of old.

Storage Systems

Pharma generates *huge* amounts of data – from trial patient records to genomic sequences. Modern datacenters include a **tiered storage architecture** to handle this variety of data. High-performance parallel file systems (for example, Lustre or GPFS) or NVMe-based storage arrays might serve active HPC scratch data where extreme I/O throughput is needed. In contrast, large-scale archival storage (petabytes of clinical data or experiment results) might reside on cost-effective distributed storage or object storage (like on-prem S3-compatible systems or cloud storage buckets).

Traditional **SAN/NAS** appliances are evolving into **software-defined storage** solutions. These allow pooling disks across many servers and presenting them as one logical storage system, managed by



datacenters are adopting data lake architectures – aggregating structured and unstructured data in one repository (often using Hadoop or cloud-based data lakes).

A key consideration is performance vs. capacity. *Genomic data* is illustrative: A whole human genome at 30× coverage can be several hundred gigabytes and takes significant time to process ^[6]. If a lab is sequencing thousands of genomes, the storage system must ingest and stream many terabytes daily. Similarly, cutting-edge imaging like cryo-electron microscopy (cryo-EM) produces **1–10 terabytes of raw data per experiment** ^[7]. The storage layer in a modern pharma datacenter is thus optimized for both high I/O (using SSDs, NVMe, and parallel access for active data) and high capacity (using dense disk drives or cloud object storage for long-term data). Data **replication and backup** are also crucial – regulatory guidelines often require retaining research and clinical data for many years, so modern datacenters integrate backup appliances, replication to remote sites or cloud, and even tape libraries for cold storage of immutable backups.

Networking and Connectivity

The "circulatory system" of the datacenter is the network. In modern designs, traditional hierarchical networking (core-aggregation-access) is giving way to flatter **leaf-spine architectures** using high-bandwidth switches (10/40/100 GbE and beyond). This ensures low-latency, high-throughput connections between compute nodes, storage systems, and gateways to the outside. For HPC clusters, ultra-low latency fabrics like InfiniBand (40G, 100G HDR etc.) or NVIDIA's **NVLink** (for GPU direct communication) are often employed, as they dramatically speed up parallel processing tasks.

Pharma companies also need robust **external connectivity**. Their datacenters connect to remote research sites, manufacturing plants, CRO partners, and cloud providers. Many use dedicated fiber links or MPLS networks for reliable high-speed data transfer (for instance, shipping clinical data from trial sites to the central datacenter). Modern datacenters incorporate **SD-WAN** technologies to intelligently route traffic over the best path, improving performance for globally dispersed operations.

Software-Defined Networking (SDN) is another component – allowing programmatic control of network flows and segmentation. In a pharma context, SDN can isolate sensitive data flows (e.g., separating a clinical trial subnet from the general corporate network) and apply dynamic security policies. This is useful for implementing Zero Trust security and micro-segmentation, which are increasingly considered best practice to prevent lateral movement of threats. In short, the modern



Virtualization and Hyperconvergence

Virtualization underpins much of the modern datacenter's flexibility. By abstracting physical hardware into virtual machines (VMs), IT teams can provision new servers in minutes (as VMs) without waiting for new hardware. This is extremely useful in pharma R&D where scientists might need a new analysis server or a specific software environment quickly. Virtualization also aids **resource pooling** – many pharma datacenters report average server utilization jumping from under 20% to 70%+ thanks to virtualization consolidating workloads. Hypervisors (like VMware ESXi, Hyper-V, or KVM) are standard in today's on-prem environments.

Building on virtualization, **hyper-converged infrastructure (HCI)** has gained popularity. HCI appliances combine compute + storage + virtualization in a single modular unit, which can be clustered with others. This simplifies management (one admin interface controls VMs *and* storage) and scales easily by adding nodes. Pharma companies are using HCI for remote sites or smaller datacenters, and even in core datacenters for certain workloads. As noted earlier, migrating to HCI helped one pharma company drastically reduce their datacenter size and power usage (^[31]). Leading HCI solutions (e.g., Dell VxRail, Nutanix, HPE SimpliVity) support the performance and fault-tolerance needs of enterprise and GxP systems, making them suitable for validated environments.

Containerization and Orchestration Platforms

While virtualization deals with VMs, **containerization** operates at the application level, packaging apps and their dependencies into lightweight containers (e.g. using Docker). Containers are highly portable between environments and incur less overhead than full VMs. In pharma IT, containerization is transformative for both dev/test and production: researchers can containerize an analytic pipeline or an AI model training job and run it consistently on a laptop, an on-prem cluster, or a cloud Kubernetes service. It greatly eases reproducibility – crucial for scientific workflows.

The rise of **microservices architectures** in software also aligns with container use. For instance, a laboratory information system might be broken into microservices (for data ingestion, processing, reporting) that run as containers, enabling independent scaling and updates. To manage tens or hundreds of containers, orchestration is needed – which is where **Kubernetes** and similar platforms come in. Kubernetes (K8s) has become a staple of the modern datacenter, orchestrating container deployment, scaling, load-balancing, and self-healing. Pharma companies use Kubernetes to



containerized tasks) can be executed in a cloud-native way rather than via traditional batch schedulers. Orchestration platforms also make it easier to adopt a **hybrid cloud** model – containers can be moved or burst to cloud clusters if on-prem capacity is insufficient.

In addition to Kubernetes, workflow orchestrators and pipeline managers (like Apache Airflow or Nextflow) are used to coordinate complex multi-step processes such as genomic analysis pipelines. These tools often run on top of the container infrastructure, ensuring that each stage of an analysis (data prep, computation, aggregation) runs in the right sequence and environment. The net benefit of containerization and orchestration is agility: pharma IT can roll out updates faster (since container images can be updated and redeployed continuously), ensure consistency across environments, and better utilize resources by packing containers efficiently on servers.

Cloud, On-Premises, and Hybrid Architectures in Pharma

The pharma industry today leverages a mix of **cloud, on-premises, and hybrid datacenter architectures**. Each approach has its pros and cons for different use cases. Below is a comparison outlining their characteristics for pharma:

On-Premises Datacenters: These are privately owned and operated facilities, traditionally the mainstay of pharma IT. **Pros:** Complete control over data and systems (which appeals in a regulated setting), low latency to local users, and often easier validation since the environment is static. On-prem can be cost-effective at steady large scale (no ongoing cloud fees) and can be tailored exactly to a company's needs (special hardware, custom security measures). **Cons:** High upfront **CapEx** – companies must invest in servers, storage, networking gear, and physical building costs (power/cooling). They also incur higher **OpEx** for maintenance, power, cooling, and staffing to manage the infrastructure ([8]) ([9]). Scaling up is slow – adding capacity could take weeks or months of procurement and installation. There's also risk of over-provisioning (wasting money on underutilized hardware during lulls) or under-provisioning (running out of capacity during spikes). In the fast-moving pharma landscape (e.g. sudden need to process vaccine trial data), this lack of agility can be problematic ([10]).

From a **compliance** perspective, on-prem gives a sense of direct control – data never leaves the company's facilities. However, maintaining compliance is entirely the company's burden: on-prem



on-prem, which can make it costly over the long term ([11]).

Cloud Architectures: This refers to using public cloud services (like AWS, Azure, Google Cloud) for infrastructure. **Pros:** High **scalability and flexibility** – resources can be expanded or reduced on-demand and paid as operating expense. Pharma companies can spin up massive compute farms for a genomics project and spin them down after, only paying for what was used. Cloud also enables global access and collaboration; employees and researchers can securely access data from anywhere, which enhances multi-site projects and partnerships. Another benefit is offloading maintenance – cloud providers handle hardware refresh, facility operations, and often routine patches/upgrades, meaning pharma IT teams can focus more on applications and data than on racking servers. Cloud services are also continuously innovating, offering cutting-edge tools (AI/ML services, data analytics platforms, managed databases, etc.) which pharma can leverage without reinventing the wheel. Given the rapid growth of AI workloads, global AI spending is projected to reach \$1.5 trillion in 2025 and exceed \$2 trillion by 2026, fueling demand for high-performance chips, data centers, and widespread applications ([12]). The global pharmaceutical industry's investment in AI specifically is estimated to reach \$2.51 billion in 2026 and grow to \$16.49 billion by 2034 ([13]) – a trend that pharma is actively driving with AI-powered drug discovery and clinical operations. According to a Google Cloud survey in late 2025, 44% of healthcare and life sciences executives reported their organizations were actively using AI agents, with 34% deploying 10 or more agents across their operations.

Cost: Cloud's pay-as-you-go model usually means lower upfront cost (no need to buy hardware) and potentially lower TCO for variable workloads. In fact, companies can save an average of 43% in infrastructure costs by moving to cloud, according to one estimate ([14]). However, cloud costs need to be monitored – at large scale or with always-on workloads, cloud can sometimes become *more* expensive than on-prem if not optimized (e.g., paying for data egress or idle resources). Still, for many pharma scenarios (bursty computations, seasonal trials, etc.), the cost flexibility is a huge advantage.

Compliance & Security: Leading cloud providers have invested heavily in meeting industry compliance standards and offer built-in security features. For example, providers offer pre-configured environments compliant with **HIPAA** and **GxP** regulations (with features like encryption, access controls, and audit logging out-of-the-box) ([15]). AWS, Azure, and GCP each have documentation and certifiable services for HIPAA, and while there's no official "FDA certification" for cloud, they align with regulations like 21 CFR Part 11 by providing the necessary controls. This can reduce the burden on pharma companies – they can inherit a secure infrastructure baseline. As



providers undergo independent audits (SOC, ISO 27001, FedRAMP, etc.) that pharma firms can leverage as part of their vendor qualification. On the flip side, ultimate responsibility still lies with the pharma company to **configure and use** cloud services properly – misconfigurations can lead to breaches. Indeed, as of 2023, *82% of data breaches involved data stored in the cloud* (public or private) and 39% of those involved multiple cloud environments (^[16]). This reflects that broad cloud adoption brings security challenges, and organizations must implement strong cloud security practices. Cloud providers do offer many native security tools (identity management, key management, threat detection) that, if used well, can make cloud very secure.

Hybrid Architectures: A hybrid approach combines on-prem and cloud, aiming to get the best of both. Many pharma companies have adopted a hybrid cloud strategy. **Pros:** Allows keeping sensitive or highly regulated data and workloads on-prem (for example, a clinical trial database subject to 21 CFR Part 11 can remain in a validated on-prem system), while offloading other workloads to cloud (like running an AI analysis on de-identified data in AWS). Hybrid can provide a **burst capacity**: the on-prem cluster handles baseline load, but when demand spikes, additional workloads "burst" into the cloud to use elastic resources. This ensures high performance without maintaining excessive hardware year-round. It also adds resilience – if one environment fails or is constrained, work can shift to the other. **Cons:** Complexity is higher. Managing a hybrid cloud means integrating disparate environments, which involves consistent networking (often via VPNs or dedicated links), unified identity and access management across on-prem and cloud, and ensuring data is synchronized or accessible in both. There can be latency when moving large datasets between on-prem and cloud, so decisions on data locality are important (some pharma firms pre-stage large datasets in the cloud if they intend to do cloud analytics, etc.). Compliance can also be complex – processes must ensure that when data moves to cloud, it is still handled under required controls (e.g., audit trails maintained).

Despite the challenges, hybrid architectures are very appealing to pharma. They allow a gradual cloud adoption (move suitable workloads to cloud, keep others local), avoid vendor lock-in to an extent, and support a transition phase where legacy systems can be modernized stepwise. Many vendors provide hybrid cloud solutions: e.g., **Azure Stack** or AWS Outposts bring cloud-like infrastructure on-prem, and container orchestration tools (like Kubernetes) can span on-prem and cloud clusters to give a unified platform. With careful architecture, a hybrid model lets pharma companies treat cloud as an extension of their datacenter. For instance, a pharma IT team might run a Kubernetes cluster where some nodes are on-prem VMs and others are cloud instances, all managed together – researchers submit jobs and the scheduler decides where to run it based on policies (on-prem if possible, else cloud). This level of integration is increasingly common.



them together. Many pharma CIOs now pursue a "cloud-smart" strategy (using cloud where it makes sense, not blindly for everything) and invest in hybrid cloud management tools. Industry analysts predict that by 2028, cloud will have shifted from being a disruptive option to simply *business necessity* – indeed, this transition is already well underway as more than 85% of biopharma executives surveyed said they would invest in data, AI and digital tools in 2025 to build supply chain resiliency, and 90% are investing in smart manufacturing for increased efficiency ⁽¹⁷⁾. Below is a simple comparison table of pros/cons:

ARCHITECTURE	PROS	CONS
On-Premises	Full control; low latency; fixed predictable cost at scale; easier physical oversight and isolation.	High upfront CapEx; long maintenance & compliance under/overutilization.
Cloud	On-demand scalability; pay-as-you-go; latest technology services available; lower IT maintenance; global accessibility; built-in compliance options.	Ongoing OpEx can grow; strong governance to prevent misconfiguration risks; vendor lock-in
Hybrid	Flexibility to optimize placement; can meet compliance by keeping regulated pieces on-prem; burst capacity and resilience; avoids all-or-nothing decisions.	Added complexity in network management; requires skilled staff and tooling to manage potential latency moving workloads

For most pharma organizations, the end-state is a hybrid cloud datacenter, with a thoughtful distribution of workloads based on regulatory, performance, and cost considerations.

Supporting Pharma-Specific Workloads

Modern datacenters are designed to support the **unique workloads of the pharma and life sciences industry**. These workloads often push the limits of data volume, computation, and compliance. Let's look at key categories of pharma workloads and how datacenter infrastructure supports them:

Clinical Trials Data Management



must provide a reliable and compliant environment for **Electronic Data Capture (EDC)** systems and clinical databases. This includes highly available database servers (often running on clustered VMs or cloud database services) to ensure investigators can input and query data 24/7.

Clinical data is sensitive (personally identifiable health information) and thus often falls under **HIPAA** and **GDPR** regulations. The datacenter must enforce strict access controls and encryption for data at rest and in transit. Many pharma companies now use cloud-based EDC or clinical data management platforms, meaning the datacenter extends to the cloud provider hosting those systems – connectivity and trust with that cloud environment is critical. For on-prem setups, the infrastructure often includes Citrix or VDI solutions to allow researchers secure remote access to trial data without exposing it directly (especially important during pandemic times for remote trial monitoring).

Modern datacenters also facilitate **clinical data integration** – pulling in data from wearables, electronic health records, and laboratory systems into a centralized data platform. This can involve streaming data pipelines and big data platforms operating in the datacenter. For example, an architecture might use Apache Kafka (for data streaming) feeding into a data lake (storage tier) and then use cloud analytics services for insight generation. The underlying compute for these tasks might be containerized microservices that can scale out as more data streams in. The key is that the modern infrastructure can handle *velocity and variety* of data typical in modern trials (including unstructured data like patient scans or doctor's notes).

Finally, **archiving and traceability** are crucial: Every modification to clinical data must be logged (audit trails as per 21 CFR Part 11), and data must be retained for years for FDA inspections. Modern storage solutions in the datacenter are configured to retain point-in-time snapshots or immutable logs to support this. In practice, pharma datacenters may integrate specialized compliance software that monitors and logs all database changes for regulatory records (^[18]).

Genomic Research and Bioinformatics

Genomics and bioinformatics are now core to drug discovery and precision medicine. These workloads are extremely data- and compute-intensive. A single human genome sequence (raw) can be 100+ GB, and processing it (alignment, variant calling) can consume 30 CPU-hours or more (^[6]). Multiply that by thousands of genomes in a population study – it's clear HPC infrastructure is needed.



scheduler (Slurm, PBS Pro, or Nextflow for workflow management) will manage these tasks, queuing up analysis of each sample to run on available nodes. Modern clusters often use **containerized pipelines** (each analysis step is a container with the bioinformatics tool, e.g., GATK) to ensure consistency and easy scaling on any node.

Storage is also tailored for genomics: high-throughput parallel file systems for active analysis (since reading/writing BAM files and intermediate data quickly is key), and large object storage for archival of raw reads and variant files. The ability to scale storage and compute jointly is important – some pharma companies leverage cloud for the largest crunches, such as national genomics initiatives or when needing to collaborate across institutions (cloud facilitates sharing data in a controlled manner). In fact, cloud providers have specialized services (e.g., Google Cloud Life Sciences, AWS Genomics CLI) to streamline genomic workflows on their infrastructure.

Another aspect is **proteomics and other 'omics** data – similarly heavy, requiring HPC and high throughput. Cryo-EM imaging mentioned earlier falls here too: converting raw microscopy images into 3D molecular structures is computationally intensive, benefiting from GPU acceleration and large-memory nodes. Modern datacenters incorporate GPU clusters for these tasks; a pharma research team might use a cluster of GPU-accelerated nodes for cryo-EM analysis or AI-driven protein folding. Notably, Otsuka Pharmaceutical used an AWS HPC setup with services like AWS ParallelCluster and Amazon SageMaker to accelerate drug discovery research – tasks like cryo-EM single particle analysis that took **7 days on-prem now complete in ~2 days on cloud HPC** (^[19]) (^[20]). This showcases how combining scalable compute and optimized hardware (cloud or on-prem) supports advanced research workloads.

AI/ML Pipelines and Computational Drug Discovery

AI and machine learning have become indispensable in pharma, from identifying drug targets to optimizing marketing strategies. Datacenters have evolved to run **AI/ML pipelines** efficiently. A typical pipeline might involve data preprocessing, model training (which can take days on large GPU clusters for deep learning models), and then deployment of models for inference. For example, training a deep neural network to analyze pathology images for toxicity signals requires reading millions of images – high-speed storage and powerful GPUs (or TPU accelerators) are needed.

Modern pharma datacenters often include dedicated AI infrastructure: **GPU farms** (e.g., servers with 4, 8 or more GPUs, possibly connected via NVLink), sometimes even specialized hardware like **TPUs (Tensor Processing Units)** or **FPGAs** for certain workloads. NVIDIA's DGX systems or similar



offers P3/P4 instances with NVIDIA A100 GPUs) or use Azure's ML service with distributed training. This elasticity is crucial when AI projects scale up.

Additionally, pharmaceutical AI often involves **natural language processing** (NLP) on scientific literature or patent data, and more recently, generative AI for molecule design (e.g., using transformers to generate candidate drug molecules). These are cutting-edge and computationally heavy. Modern datacenters must accommodate the frameworks used (TensorFlow, PyTorch, JAX, etc.) which are typically run in container environments for portability. To manage many experiments, teams leverage **orchestration** (Kubernetes with Kubeflow for ML, or Horovod for distributed training) to schedule AI jobs across the cluster or cloud instances.

Once models are trained, deploying them also relies on the datacenter. For instance, an AI model that predicts chemical compound properties might be deployed as a microservice within the datacenter (in a container on Kubernetes) so that chemists can query it from a web interface. That requires integration with the networking and security layers to expose an API while restricting access appropriately. The modern datacenter provides the sandbox for these AI inference services, ensuring they are reliably hosted and scaled when demand increases.

A notable trend is collaboration between pharma and tech companies in AI. In 2026, the industry is seeing unprecedented partnerships: the \$1 billion NVIDIA-Eli Lilly co-innovation lab announced in January 2026 (built on NVIDIA's Vera Rubin architecture with over 1,016 Blackwell Ultra GPUs rated at more than 9 exaflops), and collaborations between NVIDIA and companies like Thermo Fisher to build autonomous lab infrastructure for scalable scientific discovery. Chai Discovery, Basecamp Research, and Boltz are connecting NVIDIA BioNeMo with agentic AI and physical AI to scale drug discovery (^[4]). The datacenter (on-prem or in cloud) is where these frameworks run, so it must be equipped accordingly.

Regulatory Data Repositories and Analytics

Pharma is a highly regulated industry, and vast amounts of data must be stored and readily retrievable for compliance, audits, and regulatory submissions. This includes **quality and manufacturing data** (e.g., batch records, equipment logs), **pharmacovigilance data** (adverse event databases), and submission archives (electronic common technical documents, eCTDs). Modern datacenters serve as the backend for validated systems managing this information.



infrastructure with strict controls. The datacenter often uses redundant architecture (clustered databases, high-availability storage, robust backup) to ensure no data loss. Features like **WORM (Write Once Read Many)** storage or append-only logs may be used for audit trail data to prevent tampering. As per FDA 21 CFR Part 11, any electronic record system must have secure, computer-generated time-stamped audit trails that record operator actions and changes ([21 CFR Part 11 Audit Trail Requirements \[Explained\] - SimplerQMS](#)). Modern database and content management systems include such audit trail capabilities, but the infrastructure must support it (through sufficient storage and ensuring those logs are also backed up). Leading platforms now incorporate continuous validation and immutable audit trails by design so that compliance is "built-in" to the data infrastructure (^[18]).

Analytics on regulatory data is also growing. Companies want to proactively use their compliance and quality data for insights (for instance, analyzing manufacturing data to detect potential quality issues or using AI on pharmacovigilance reports to spot safety signals). This means the datacenter not only stores the data but also provides analytic platforms (like data warehouses or Hadoop/Spark clusters) that can be used without violating data integrity. Often a separate analytics environment is fed from the primary records (to avoid messing with the primary database). Modern data architecture (via data lakes and ETL pipelines) helps create these secondary datasets for analysis. Cloud services are often tapped here, since they can provide scalable analytics without risking the core validated systems – for example, exporting de-identified patient data from a clinical database into a secure cloud analytics environment for further research.

In essence, pharma-specific workloads demand that the datacenter be both **robust and nimble**: robust in handling huge data and strict compliance, nimble in scaling up compute for heavy science and enabling new technologies like AI. The modern pharma datacenter achieves this through a combination of HPC design principles, cloud integration, and adopting best-of-breed tools for data management.

Regulatory Compliance in the Modern Datacenter (HIPAA, GxP, 21 CFR Part 11)

Compliance is paramount in pharmaceutical IT. Any modern datacenter architecture must enable and not hinder compliance with regulations such as **HIPAA, GxP, and 21 CFR Part 11**. Let's break down these requirements and how datacenter practices address them:



administrative, physical, and technical safeguards for any system handling patient data (e.g., clinical trial participant info or patient registries in pharma studies). **Technical safeguards** include access controls (unique user IDs, session timeouts), encryption of data at rest and in transit, and audit controls to record accesses to sensitive data. Modern datacenters enforce these via centralized identity and access management (IAM) systems, role-based access control, and encryption services. For example, many pharma IT setups use encryption on databases and file systems containing PHI, and manage keys using secure key management appliances or cloud KMS (Key Management Service). On the physical side, the data center (whether on-prem or at a cloud provider) must have strong security – restricted access, surveillance, etc., which is standard in enterprise datacenters. Cloud providers also assert compliance with HIPAA by signing Business Associate Agreements (BAAs) and having many **in-scope services** that have the necessary safeguards ([21]). A pharma company can confidently use those services but must still configure them correctly (e.g., ensure a cloud storage bucket with PHI isn't left public).

- **GxP:** This is an umbrella term for "Good [Manufacturing/Laboratory/Clinical] Practices" – regulations ensuring product quality and patient safety. In IT terms, when GxP processes use computerized systems, those systems must be **validated** and managed under strict change control. FDA's 21 CFR Part 11 is actually one part of GxP focused on electronic records. Modern datacenters facilitate GxP compliance by providing **validated infrastructure**. This could mean using qualified vendors and equipment, maintaining IQ/OQ/PQ (installation/operational/performance qualification) documents for infrastructure, and controlling any changes to the environment. A significant development is the FDA's **Computer Software Assurance (CSA)** draft guidance, first issued in September 2022 and expected to be finalized by late 2025, which aims to replace traditional Computer System Validation (CSV) with a streamlined, risk-based approach. Under CSA, validation efforts are scaled to the risk a software poses to quality – high-risk functions still require rigorous scripted testing, while low-risk tools can be qualified with supplier documentation and unscripted testing ([22]). Some pharma companies maintain separate "validated" environments in their datacenter for GxP applications, segregated from non-GxP workloads. However, with virtualization and cloud, this line is blurring. FDA guidance allows cloud and virtualization for GxP systems provided you validate their intended use and maintain control. For instance, AWS has published guidance on using AWS for GxP, noting that while AWS doesn't get "certified" for GxP (no such certification exists), many pharma companies have successfully qualified AWS services as part of GxP systems ([23]).



redundant storage (preventing data loss from hardware failure), automated backup and disaster recovery, and **audit trails**. Part 11 requires audit trails for any creation/modification of electronic records used in GxP activities (^[24]). Modern databases and applications often have built-in audit trail features, but the IT team must ensure these are enabled and that audit logs are stored securely (often on separate, append-only storage). Technologies like blockchain have even been explored for immutable audit trails, but more common is simply using database logging with strict access. Additionally, **electronic signatures** are addressed by Part 11 – systems must uniquely identify users and bind their signature (authentication) to records. This means datacenter IAM integration, enforcing unique credentials, and time stamps via server clocks (often synced with NTP). Pharma datacenters typically integrate with corporate Active Directory or similar for unified identity, which helps meet this requirement by ensuring each action is tied to an authenticated user account.

Another practice is maintaining **validation documentation** for infrastructure as code. If using Infrastructure-as-Code (IaC) to deploy cloud resources or VMs, those IaC scripts and their test results can serve as evidence of proper configuration (sort of like an automated IQ/OQ). Tools and processes are emerging to continuously validate cloud configurations against compliance rules (for example, using AWS Config conformance packs for 21 CFR Part 11 controls (^[25])). In short, compliance is increasingly achieved by *designing it into the infrastructure* – strong identity, logging, and change management built into the datacenter fabric from the ground up.

- **21 CFR Part 11:** This FDA regulation specifically governs electronic records and signatures for any FDA-regulated process (covering drugs, biologics, medical devices, etc.). In summary, Part 11 requires system validation, the ability to generate accurate copies of records, protection of records, limited system access, audit trails, operational system checks, authority checks, and training of users, among other items. Modern datacenters meet these in various ways:
- **System validation:** All critical software (infrastructure software and applications) must be validated to do what it's intended. For IT infrastructure, this might mean qualification of the platform (e.g., validating that a cloud VM environment properly provisions consistent VMs, or that a storage system reliably stores/retrieves data). Pharma companies often leverage vendor validation packages or do in-house testing to qualify new datacenter tech. The use of **automation** helps – for instance, using test suites to verify a new Kubernetes cluster deployment meets all security and performance specs before using it for GxP data.
- **Record retention and retrieval:** The datacenter must ensure that records are retained for the required period (often years or decades) and can be readily retrieved in human-readable form for FDA inspection. This is addressed by robust archival storage and database maintenance. Many pharma firms use write-once storage or backup to cloud archives to ensure long-term



- **Security and access controls:** Part 11 dovetails with cybersecurity best practices – requiring unique user accounts, password policies, and controls to **discourage unauthorized access**. Modern datacenters support this with enterprise IAM solutions, multi-factor authentication for remote access, network segmentation (so that only authorized segments can reach certain servers), and monitoring tools that can alert on any anomalous access attempt. In fact, **zero trust architecture** is being adopted to further tighten access – assuming no user or device is trusted by default and continuously verifying credentials and context.
- **Audit trails and change control:** We discussed audit trails – the datacenter should not only keep application-level audit logs but also infrastructure logs (e.g., who logged into a server, who changed a firewall rule, etc.). Part 11 expects that any changes to records are recorded. Modern log management solutions (like SIEM systems) aggregate logs from all components, providing a central audit repository that can be queried during compliance checks. Change control is facilitated by configuration management databases and IT service management processes; for instance, no update is applied to a GxP server without an approved change request, and the datacenter automation will only execute changes that have proper approval tags. This process can be partly automated through pipeline approvals in Infrastructure-as-Code deployments.

In summary, compliance is "baked into" the modern pharma datacenter via a combination of technology and process. Technologies provide the capabilities (encryption, logging, identity management, redundancy), while processes and validation efforts ensure those capabilities are correctly implemented according to regulatory expectations. As one industry insight put it, for pharma applications **compliance must be built into the analytics infrastructure from the ground up**, with continuous validation and immutable audit trails ^[18]. Modern datacenters are enabling this by providing more automated, monitored, and secure environments than ever before, which helps companies meet HIPAA and FDA requirements without completely sacrificing agility.

Data Security and Privacy Challenges

With great data comes great responsibility – pharma datacenters hold some of the most sensitive data (patient health info, proprietary research, drug formulas), making security and privacy a top concern. Modern datacenters face evolving **threats** and must rise to meet them. Here are key challenges and how they're addressed:

- **Cybersecurity Threats:** Pharma has been a prime target for cyber attacks (espionage by nation-states looking for IP, ransomware gangs seeking payout, hacktivists, etc.). A breach can



\$5.10 million in 2024), while **healthcare breaches remain the costliest at \$7.42 million** (^[26]) – but also erode public trust and compromise patient privacy. Healthcare data breaches take the longest to identify and contain at an average of 279 days, five weeks longer than the global average breach lifecycle. One major challenge is the sheer **attack surface** – modern datacenters are hybrid and accessible globally, which while beneficial, also means more entry points for attackers. Phishing was the leading initial access vector in 2025, accounting for almost 16% of data breaches, replacing stolen credentials (10%) from the previous year (^[27]). Studies continue to show that human error remains a primary cause of breaches, indicating the need for strong user training and technical controls to mitigate mistakes.

- **Access Control and Insider Threats:** Controlling who can access what data is harder as datasets grow and more collaborators (internal or external) need access. The principle of least privilege must be enforced via robust IAM policies. Modern datacenters integrate identity across on-prem and cloud, often tying into federated identity systems. Technologies like **Privileged Access Management (PAM)** are deployed to secure admin accounts (which could do the most damage in a breach). Insider threats (disgruntled employees or careless users) are mitigated by monitoring and by compartmentalizing data. For example, a research scientist's account should only have access to the projects they work on, not everything. Role-based and attribute-based access controls (RBAC/ABAC) help achieve granular policies. Additionally, **data masking and tokenization** can be used so that even if a user has access to a dataset, identifying details are obscured unless they have a need to see them. Many clinical systems will separate personal identifiers from clinical data, replacing, say, names with codes – the mapping file is kept in a highly secure enclave.
- **Network Security and Segmentation:** Modern datacenter networks are segmented by design (using VLANs, SDN, or cloud VPC segmentation). Sensitive systems (like a regulatory submissions database) might live on a network segment that has no direct internet access and only a few jump servers can reach it. Firewalls – both at the perimeter and internally – guard these segments. An emerging best practice is **micro-segmentation**, which can be done via SDN or host-based firewalls: each application or service is isolated even from others, so if malware infects one server, it can't freely spread. This helps contain breaches; given that in 2023, once intruders got in, they could move laterally within **84 seconds** on average (^[28]), segmentation is critical to slow them down and allow detection. Modern datacenters also often employ **intrusion detection/prevention systems (IDS/IPS)** and continuous network monitoring (using AI to spot unusual traffic patterns that might indicate a breach in progress).



storage containing sensitive data – using tools like transparent data encryption for databases or self-encrypting drives. In cloud, it's standard to enable volume and object storage encryption (with customer-managed keys where possible). **Encryption in transit** is enforced by using TLS for all connections; for internal API calls or service-to-service communication, service mesh technologies (like Istio in Kubernetes) can enforce TLS everywhere. Even if data is stolen, encryption can keep it safe (assuming keys are well protected). Some are looking at **homomorphic encryption** or secure enclaves to allow computing on encrypted data (useful for collaborative research where data can't be fully shared), but these are still niche in deployment.

- **Security Monitoring and Incident Response:** A modern datacenter is never static – hence continuous monitoring is needed. Security Information and Event Management (SIEM) systems aggregate logs from servers, network devices, and cloud services to flag suspicious events. For example, multiple failed login attempts on a database or an admin account logging in at odd hours would trigger alerts. Many organizations also employ **security operations centers (SOCs)** and advanced tools with machine learning to detect anomalies. Given the earlier statistic that a large share of breaches involve cloud components (^[29]), monitoring cloud configurations is essential too – using cloud security posture management (CSPM) tools to catch misconfigured storage or overly exposed services before attackers do. Modern datacenters support this by providing APIs and integration points for all components so they can be monitored centrally.

When an incident is detected, a robust response plan is required. This includes having backups (offline backups especially, to recover from ransomware) and practicing disaster recovery. It's worth noting that **high availability and DR** are not just for uptime – they are also a security measure (ransomware can be mitigated if you can wipe systems and restore clean data quickly). The case study earlier noted the pharma co. had high availability and DR in place after modernization. Many pharma companies perform periodic DR drills and also test their incident response to cyber scenarios, often mandated by internal policies or regulations.

- **Privacy Compliance:** Beyond technical security, regulatory privacy requirements (like GDPR, California Consumer Privacy Act, etc.) impose obligations on handling personal data. Pharma often deals with patient data from global trials, so compliance with GDPR (for EU data) is big. The datacenter must enable capabilities like data localization (keeping EU data within EU data centers if needed), data deletion (if a subject withdraws consent, all their data must be deletable), and auditing of data usage. These requirements often influence architecture – for



- **Third-Party and Supply Chain Security:** Modern datacenters are interconnected with third-party systems – SaaS providers, CRO systems, etc. Each connection is a potential risk if not managed. Ensuring secure APIs/VPNs, contractually requiring security controls of partners, and even running periodic security assessments of key vendors is now common practice. When using cloud, understanding the **shared responsibility model** is crucial (the cloud provider secures the infrastructure, but the customer must securely configure their applications on it). Misunderstandings here have led to incidents like open S3 buckets. Pharma IT departments now often have cloud security teams or external audits to verify configurations.

In essence, **data security in modern pharma datacenters is a multi-layered effort:** perimeter defenses to keep attackers out, internal defenses to catch or contain those that get in, rigorous access and encryption to protect data, and monitoring and response capabilities to act swiftly if an incident occurs. The stakes are incredibly high – not only is patient data at risk, but even intellectual property like a drug formula must be shielded (industrial espionage is a real threat). Fortunately, modern technologies (AI-driven security analytics, automated compliance checks, etc.) and a culture of security (DevSecOps practices embedding security from development through operations) are helping pharma organizations stay ahead of threats. Still, vigilance is needed as attacks continuously evolve. As the adage goes, it's not *if* a breach will happen, but *when*, so the datacenter must be prepared to prevent, detect, and recover from any security event as part of its core mission.

High Performance Computing and Scalability Considerations

Pharma has an insatiable appetite for computation – from crunching vast chemical libraries in silico to simulating clinical trial outcomes. **High Performance Computing (HPC)** is therefore a linchpin of modern pharma datacenters. The challenge is not just raw performance, but also *scalability* – the ability to ramp up resources as demands spike. Here's how modern datacenter design addresses HPC and scalability:

HPC Infrastructure: Traditional HPC in pharma meant large on-prem clusters or supercomputers, often custom-built (think hundreds or thousands of CPU cores networked with high-speed interconnects, plus specialized storage). Those are still in play, but modern HPC can also leverage



big job comes (like running a million compound docking simulations for a drug discovery project), they burst to cloud HPC instances. Tools like AWS ParallelCluster or Azure CycleCloud allow setting up cloud-based HPC clusters with similar schedulers (SLURM, etc.) so the user experience is nearly the same. **Hybrid HPC** is becoming normal – a survey indicated a strong trend of mixing on-prem and cloud resources to handle computational peaks efficiently.

Workload Scheduling and Orchestration: HPC workloads in pharma vary – some are embarrassingly parallel (many independent tasks like sequence analyses), others are tightly coupled (like molecular dynamics simulations that need frequent communication between nodes). Modern datacenters employ sophisticated schedulers to allocate jobs to the right resources. For parallel independent tasks, schedulers can use cloud auto-scaling – e.g., spin up 100 cloud VMs to run genomics jobs in parallel and then terminate them. For tightly coupled tasks, the infrastructure has nodes with low-latency interconnect and the scheduler ensures those jobs get those nodes. Increasingly, container orchestration (Kubernetes) is being integrated with traditional HPC schedulers, or even replacing them for certain workloads, as mentioned earlier, since it can schedule containerized jobs across hybrid infrastructure. This makes scaling more cloud-native.

GPU and Accelerator Scaling: As mentioned, GPUs are crucial for many HPC/AI tasks. Scaling GPU workloads can be tricky – you need both many GPUs and fast connections between them for multi-GPU training. Datacenters handle this by deploying nodes with multiple GPUs (8+ per node) and using interconnects like NVLink or InfiniBand between nodes. Some have even started exploring **NVSwitch** for intra-node full GPU meshing. In cloud, one can now rent entire GPU clusters (for example, AWS offers p4d instances with 8 A100 GPUs each and you can cluster them using EFA, a high-speed network adapter, to create a multi-node GPU cluster). The ability to **scale out** AI training to dozens of GPUs means models that took weeks can train in days, accelerating research timelines.

A practical example in pharma HPC scaling: A company doing virtual screening of compounds can distribute the task across thousands of cores – if on-prem has 500 cores, they might extend to use 5000 cores on cloud for a weekend to finish the screening faster, then shut them down. The result is a shortened drug candidate selection process from perhaps months to weeks. Indeed, studies have confirmed that using cloud-based HPC can lead to faster drug discovery by enabling novel techniques at scale (^[30]).

Autoscaling for Enterprise Apps: Scalability isn't just about HPC; pharma datacenters also need to scale *enterprise workloads*. For instance, a pharma might launch a new patient portal or a



and deployed on orchestrators that can auto-scale horizontally (add more container replicas or VMs when load increases). This prevents downtime during critical periods (say a surge of trial investigators uploading data at deadline). Cloud load balancers and auto-scaling groups play a role when such apps are hosted in public cloud. On-prem, virtualization clusters can dynamically allocate more resources to an app VM if needed (assuming headroom exists), or nowadays even trigger provisioning of new VMs if integrated with something like vRealize Automation or OpenStack.

Disaster Recovery and Geo-Scaling: Scalability also encompasses geographic scaling – expanding to new regions. Pharma is global, so datacenters must support multi-site HPC (e.g., research teams in the US, Europe, Asia all using a common platform). Modern solutions replicate data across regions or use distributed file systems accessible globally. Cloud makes it easier by having global regions – a workload can be run in US-East and then in EU-West etc., or run jointly if latency allows. For DR, the ability to scale up in an alternate datacenter or cloud region if the primary fails is key. Many pharma datacenters have a "warm" standby environment that can be scaled up to full capacity if needed. For example, critical applications might be in Active/Active across two sites, or Active/Passive with periodic sync – if one goes down, the passive one scales up and takes over. With infrastructure as code and cloud images, spinning up an entire datacenter's worth of services in a new region can be automated to occur within hours, which is a form of rapid scalability in a crisis.

Performance Optimization: High performance isn't just about adding more nodes; it's also about optimizing each component. Modern datacenters incorporate performance monitoring tools that identify bottlenecks (CPU, memory, I/O). If a workload is I/O-bound, IT can allocate faster storage (NVMe SSD pools) or even adjust the HPC job's I/O strategy (like using local SSD scratch on each node). If it's network-bound, perhaps move the jobs to a tighter network cluster. There's also a trend of **HPC in the cloud with spot instances** (using spare capacity at lower cost) – while not guaranteed, it can provide massive compute at a fraction of cost if the application can checkpoint and handle interruptions. Pharma researchers, always budget-conscious, experiment with this for non-urgent but large-scale runs.

Lastly, consider **future scalability**: technologies like **quantum computing** are on the horizon, which could drastically change compute needs for certain problems (like molecular simulation). Modern datacenters are keeping an eye on these – some cloud providers already offer quantum computing resources (as experimental services). While not mainstream yet, it underscores that scalability



In summary, HPC and scalability in the modern pharma datacenter are achieved through a blend of on-prem power and cloud elasticity, with intelligent scheduling and use of accelerators. The infrastructure is designed to **scale out** (add more parallel resources) and **scale up** (more power per node) as needed. By doing so, it supports the rapid pace of scientific innovation – whether it's analyzing *millions of data points overnight* or launching a global app to support a new therapy. The result is that computational capability is rarely the limiting factor in pharma research today; if the science or business needs more compute, the modern datacenter can deliver it, virtually on demand.

Sustainability and Energy Efficiency Trends

Modern datacenters, including those in pharma, are not only judged on performance and reliability – there is growing emphasis on **sustainability and energy efficiency**. Large pharma companies often have corporate sustainability goals (such as reducing carbon footprint), and datacenters can be a significant contributor to energy consumption. Several trends and innovations are shaping greener datacenter design:

Energy Consumption Awareness: The energy impact of data centers is increasingly significant. According to the IEA, U.S. data centers consumed 183 terawatt-hours (TWh) of electricity in 2024, representing more than 4% of the country's total electricity consumption – roughly equivalent to the annual electricity demand of Pakistan ([31]). Globally, data centers consumed around 415 TWh in 2024, with projections indicating this could reach approximately 1,050 TWh by 2026 if current trends continue ([32]). By 2030, U.S. data center electricity consumption is projected to grow by 133% to 426 TWh. Critical power to support data centers' most important components is expected to nearly double between 2023 and 2026 to reach 96 GW globally, with AI operations alone potentially consuming over 40% of that power ([33]). Pharma datacenters, with HPC systems, are particularly power-hungry (HPC racks densely packed with CPUs/GPUs can draw tens of kW each). Companies are now closely monitoring Power Usage Effectiveness (**PUE** – ratio of total facility power to IT equipment power). An ideal PUE is 1.0 (all power goes to computing). Traditional enterprise data centers might have PUE of 1.7–2.0, but modern designs aim for ~1.2 or lower. Techniques like hot/cold aisle containment, more efficient UPS systems, and high-efficiency cooling help drive down PUE. For example, using outside air **free cooling** when the climate permits can significantly cut energy used by chillers. Some pharma datacenters have been built in cooler climates or use seasonal strategies to leverage outside air.



blocks or even immersion cooling (submerging servers in dielectric fluid). Studies show that switching from traditional air cooling to liquid can reduce facility power consumption significantly – one analysis found up to an **18% reduction in total data center power** and corresponding drop in energy costs by implementing liquid cooling on majority of the load ([341] ([351]). It also noted a 15% improvement in an energy efficiency metric (TUE) ([361]). Liquid cooling not only saves electricity (by eliminating many fans and allowing higher thermostat setpoints for air) but can also increase computational density (more compute per rack without overheating). However, it's early days – many pharma IT shops are conservative about adopting it widely until it's proven and aligns with their risk profile. That said, labs with extreme HPC (like for AI model training) are piloting these solutions.

Use of Renewable Energy: A major sustainability focus is sourcing power from renewable energy – solar, wind, etc. Cloud providers have been leading here: Amazon achieved its 100% renewable energy goal in 2023, seven years ahead of its original 2030 target. Amazon has become the largest corporate purchaser of renewable energy in the world for five years running, investing billions of dollars in over 600 renewable energy projects globally, generating enough energy to power more than 8.3 million U.S. homes ([371]). Microsoft and Google also claim high percentages of renewable energy for their data centers. For pharma companies using cloud, this means by outsourcing computing to cloud providers, they indirectly use greener energy than they might locally (depending on their local grid). Some analyses found big cloud providers are **3–5 times more energy efficient** than typical enterprise data centers, factoring both efficient hardware utilization and power sourcing. AWS claims that moving a typical workload to AWS can reduce its carbon footprint by 80% due to more efficient resource use. Pharma companies with on-prem datacenters are responding by buying green energy via power purchase agreements or installing on-site renewables. Notably, 53% of pharmaceutical firms are now committed to the United Nations' Race to Zero initiative, and capital expenditure is increasingly directed toward biomethane, renewable energy, and low-emission facilities ([171]).

Additionally, **carbon accounting** is becoming part of IT planning – calculating the carbon emissions of datacenter operations. This can influence decisions like migrating certain workloads to a region where the grid is cleaner. The regional impacts are significant: in Ireland, around 21% of the nation's electricity is used for data centers, with the IEA estimating this could rise to 32% by 2026. In the U.S. state of Virginia, data centers already consume 26% of electricity ([381]). This is leading to consumer price impacts – in the PJM electricity market stretching from Illinois to North Carolina, data centers accounted for an estimated \$9.3 billion price increase in the 2025-26 capacity market, with average residential bills expected to rise by \$16-18 per month in some areas ([391]).



Heat Reuse: A fascinating trend in sustainability is reusing the waste heat from datacenters. In some regions, datacenter heat is used to warm nearby buildings or even greenhouses. While this is more prevalent in big cloud or colocation facilities (for example, a datacenter in Denmark piping heat to a community heating system), a pharma company could conceivably partner with local facilities to reuse heat from its on-prem datacenter. This requires proximity and planning, but it turns a waste product into a resource, improving overall efficiency (some Scandinavian datacenters advertise "negative" heating costs for neighbors in winter due to their output).

Efficient Hardware Utilization: One often overlooked aspect – the **most efficient server is the one you don't run**. Modern datacenters use consolidation and smarter scheduling to ensure that idle servers are minimized. By running at higher utilization (and turning off or sleeping unused machines), you improve work-per-watt. Virtualization and containers help here by packing workloads. Some organizations now measure workload energy efficiency – e.g., tracking kilowatt-hours per research job – and optimizing accordingly. If a job can run on GPU in 1 hour vs CPU in 10 hours, even if the GPU uses more power, the total energy might be lower for the GPU case due to shorter runtime. Thus, investing in the right hardware (like GPUs or newer CPU architectures) can actually save energy if it completes tasks faster. Many pharma datacenters are refreshing older equipment sooner, not just for performance but because new servers are markedly more performance-per-watt efficient. For example, an old server might draw nearly the same power as a new one, but perform half the computations; replacing it means doubling output for the same energy.

Sustainable Facilities: The building and cooling infrastructure around the IT equipment also sees innovation. Some trends include:

- Using **evaporative cooling** (with water) which is more efficient than compressor-based cooling on dry days, though it increases water usage (a trade-off: data centers can consume millions of gallons for cooling). There's awareness of water consumption as part of sustainability – places like Arizona or Singapore are encouraging waterless cooling methods to conserve water ^[40] ^[41].
- Designing for higher ambient temperatures: ASHRAE has expanded acceptable server inlet temps, so some datacenters run a bit "hotter" (like 27°C instead of 18-20°C) to reduce cooling energy. The risk is slightly higher component wear, but if within tolerances, it's fine. It's reported that every 1°C increase in setpoint can save a few percent in cooling energy, so modern DC operators carefully tune this.



equipment level to throttle down CPUs when not under load to save power. Even using direct current (DC) distribution in some datacenters to avoid losses from AC conversions is an approach.

Cloud Impact: Many pharma companies have concluded that moving generic workloads to large cloud providers can reduce their carbon footprint, because the cloud provider operates at scale with very efficient practices and higher renewable mix. For example, AWS claims that moving a typical workload to AWS can reduce its carbon footprint by 80% due to more efficient resource use, and up to 96% after AWS transitions to 100% renewable energy ^[42]. This doesn't mean everything should go to cloud, but it's a factor when considering sustainability alongside cost and control.

End-of-life and Circular Economy: Sustainable datacenter practice also considers what happens to hardware at end-of-life. Are servers recycled responsibly? Some companies refurbish and reuse hardware for less critical tasks or donate it if still functional. Proper recycling of electronics (to reclaim materials and avoid toxic waste) is part of green IT policies.

All these trends show that pharma datacenters are aligning with broader environmental responsibility goals. A pharma IT leader today is likely tracking metrics like PUE, carbon emissions per workload, etc., whereas a decade ago they might have only cared about uptime and cost. This shift is driven by corporate social responsibility and also by practical cost savings – using less energy is both green and budget-friendly in the long run, especially as energy costs can be significant for running big datacenters. Additionally, regulators and partners are starting to ask about sustainability in the supply chain, so demonstrating that the datacenter is energy-efficient can be a plus in business terms.

In conclusion, the modern pharma datacenter strives to be *"high-performance" in an environmental sense too*. Through efficient design, innovative cooling, renewable energy, and smarter operations, it's possible to support the computational needs of pharma while minimizing environmental impact. Sustainability is now a key pillar of datacenter excellence, alongside security, scalability, and reliability.

Vendor Landscape and Ecosystem Tools

Building and running a modern datacenter is a complex endeavor, and pharma IT departments rely on a rich ecosystem of **vendors and tools**. Here we compare and highlight some of the key players and solutions relevant to pharmaceutical datacenters:



Amazon Web Services (AWS): AWS is a dominant cloud choice in pharma, in part due to its maturity and breadth of services. AWS provides global infrastructure with many compliance certifications (including support for HIPAA, GxP, etc.). Pharma companies use AWS for HPC (with services like AWS ParallelCluster to set up HPC clusters, and FSx for Lustre providing high-speed storage for HPC). AWS has specific life sciences initiatives – e.g., AWS Data Exchange for sharing healthcare data, and AWS HealthLake for storing and analyzing health data in compliant manner. For GxP, AWS offers whitepapers and guidance; while AWS doesn't guarantee compliance out-of-the-box, they highlight customers who validated GxP systems on AWS ([43]) ([44]). A notable strength of AWS is its extensive analytics and AI services: Amazon SageMaker (used by Otsuka in the case study to develop ML models securely ([19])), Amazon Comprehend Medical (for NLP on medical text), etc. AWS also focuses on hybrid solutions – e.g., AWS Outposts (on-prem AWS-managed racks) for cases where data must stay on-site but you want cloud ops. In terms of vendor support, AWS has dedicated teams for healthcare & life sciences, which many pharma firms leverage for architecture guidance.

Microsoft Azure: Azure is also very popular, especially given many pharma companies' existing use of Microsoft enterprise software. Azure's cloud offers a strong compliance portfolio – it's often noted that Azure has a broad set of compliance offerings and tools for customers to simplify meeting standards ([45]). Azure has specific solutions like Azure for Life Sciences, and offers services tailored to genomics (they have partnerships with genomics companies, and services like Azure CycleCloud for HPC workload management and Azure Machine Learning for AI). Azure is known for its **hybrid integration** – with Azure Stack allowing Azure services to run inside a customer's datacenter, and tools to connect on-prem AD with Azure AD for unified identity. For HPC, Azure offers GPU instances, InfiniBand connectivity (HB-series VMs, etc.), and recently, Azure Quantum for experimental quantum computing access. Many enterprises trust Azure for its security focus – it has multifaceted security tools like Azure Sentinel (SIEM) and Defender for Cloud, which can be useful for unified security monitoring across hybrid environments. Pharma companies that are Windows/.NET heavy often find it straightforward to migrate apps to Azure or use Azure's PaaS databases and services.

Google Cloud Platform (GCP): GCP is the third major cloud player. It historically focused on data and analytics strengths (after all, Google's internal tech underpins it). For pharma, Google Cloud's life sciences offerings have evolved significantly – the **Cloud Life Sciences API** was deprecated in 2025, with workloads now supported by Google Cloud Batch for pipeline execution ([46]). GCP continues to offer robust genomics capabilities: users can ingest raw sequences with genome-wide association study (GWAS) pipelines into Cloud Storage, extract variants using Batch API, and



TensorFlow originated from Google, and they offer TPUs for fast AI computations. In 2025, Google introduced breakthrough tools including **DeepPolisher** for genome assembly accuracy, **DeepSomatic** for cancer-related mutation detection, and **AlphaGenome** for predicting how DNA variants affect biological processes ^[47]. According to a Google Cloud survey in October 2025, 44% of healthcare and life sciences executives reported actively using AI agents. GCP's BigQuery (a serverless data warehouse) is attractive for storing and querying large datasets with ease and speed. In compliance, GCP can sign BAAs for HIPAA and has documentation on 21 CFR Part 11 compliance. GCP received the IDC 2023 Cloud CSAT Award for Life Sciences, achieving the highest customer satisfaction scores. One advantage with Google is its emphasis on open-source and portability – tools like Kubernetes (Google originated it) help avoid lock-in. Google also touts its carbon-neutral operations and leadership in renewable energy usage, which aligns with pharma's sustainability goals.

In summary, all three major CSPs – AWS, Azure, GCP – have robust offerings for compute, storage, AI, and compliance. Many pharma companies adopt a **multi-cloud** approach, using more than one provider to avoid dependency and to leverage specific strengths. For instance, Moderna is explicitly multi-cloud, using AWS for core data storage but GCP for analytics and BI; Johnson & Johnson has a multi-cloud approach, using "the right cloud for the right workload." In December 2025, an AWS-Google Cloud multicloud partnership was announced, enabling customers to connect the two platforms more seamlessly through AWS Interconnect and Google Cloud's Cross-Cloud Interconnect ^[48]. For HIPAA compliance, all three providers can sign Business Associate Agreements, but organizations remain responsible for properly configuring and using cloud services – there is no official HIPAA certification that a CSP can obtain. Cost and existing relationships often influence provider choice. Importantly, these vendors also have marketplaces and partner ecosystems where third-party solutions (including specialized pharma software) can be deployed easily in their cloud.

On-Premises Infrastructure Vendors

Dell Technologies: Dell (including the legacy EMC storage business) is a major supplier of on-prem hardware to pharma datacenters. Dell offers a range of servers (PowerEdge line) that power compute clusters and virtualization farms. They also provide storage solutions like PowerStore, Isilon/PowerScale (for scale-out NAS often used in research labs for large file shares), and Unity/PowerMax SANs for transactional workloads. For HPC, Dell has **validated designs for Life Sciences HPC** – e.g., pre-configured clusters specifically tuned for genomics and molecular dynamics with the right balance of compute, GPU, and storage ^[49] ^[50]. Dell's **Ready Solutions for**



integrates with VMware, which many IT shops use). In networking, Dell's switches (Force10 line) are used in some datacenters as cost-effective leaf-spine building blocks. Pharma companies often have long-term support contracts with Dell, and Dell's ProSupport is known for its responsiveness, which is crucial for mission-critical systems.

Hewlett Packard Enterprise (HPE): HPE is another key player – their ProLiant servers and blade systems are widely used. HPE also acquired Cray in 2019, which means HPE can offer **supercomputing-grade systems** to pharma (Cray's technology is used in some top research labs and could benefit pharma HPC for tasks like large-scale simulations). HPE's storage portfolio (3PAR/Primera, Nimble, etc.) and newer Alletra line provides high-performance storage for different needs. HPE is focusing on an **everything-as-a-service** model with HPE GreenLake, which allows pharma companies to consume on-prem hardware in a cloud-like subscription/utility model. This is attractive if capital expense is an issue or if flexibility is needed (HPE essentially installs gear on-prem but charges based on usage). For HPC and AI, HPE offers systems like the Apollo series (dense compute for HPC) and partnerships with NVIDIA (e.g., HPE sells NVIDIA DGX systems as well). HPE also has a strong services arm that can assist with digital transformation projects, including datacenter modernization and cloud integration.

IBM and Others: Although not explicitly listed in the question, it's worth noting IBM has a presence in pharma – historically via IBM Power systems (some pharma research used IBM Power for certain HPC workloads or database servers) and storage solutions (IBM Spectrum Scale aka GPFS is a popular parallel file system in research). IBM also has a cloud (IBM Cloud) and specialized offerings like IBM Watson Health (though parts of that have been divested). However, IBM's footprint in modern cloud era has reduced relative to the big three CSPs. **Lenovo** is another server vendor that, after acquiring IBM's x86 server business, sells a lot into HPC (Lenovo Neptune is their liquid cooling HPC line, for example). Pharma companies selecting hardware often evaluate both Dell and HPE, and sometimes Lenovo for cost or specific features.

Networking Vendors (Cisco, Juniper, etc.): The physical network in many datacenters relies on Cisco – their Nexus switches are common for core and distribution. Cisco's UCS servers also combine compute/network and might be used for certain workloads (like UCS blade chassis for virtualization clusters). Cisco also provides ACI (Application Centric Infrastructure) which some pharma IT use for software-defined networking in the datacenter. Other vendors include Arista Networks (high performance switches favored in trading and increasingly in HPC due to ultra-low latency), and Juniper for routing and some switching. The choice comes down to performance, existing expertise, and sometimes bundled deals.



VMware: In on-prem environments, VMware's software stack is nearly ubiquitous. VMware vSphere runs the virtualization for countless pharma servers. On top of that, VMware's vSAN (for HCI storage) and NSX (for SDN and micro-segmentation) are often deployed to create a fully software-defined datacenter. VMware also offers **VMware Cloud Foundation** which can extend one's VMware environment to public cloud or manage across hybrid, something pharma might use to seamlessly move VMs to VMware-on-AWS for example. Given the strong validation and familiarity, many GxP systems run on VMware VMs and are expected to continue so even as cloud grows (some opt to use VMware Cloud on AWS/Azure to basically run a VMware cluster on cloud hardware, easing migration while staying in validated configurations).

Kubernetes Ecosystem: We touched on Kubernetes – many vendors offer enterprise Kubernetes distributions (Red Hat OpenShift, VMware Tanzu, Azure AKS on-prem (Arc-enabled), etc.). Pharma firms often choose one of these to simplify container management. Red Hat OpenShift is popular in regulated industries due to Red Hat's support and additional tooling (it provides a more controlled environment on OpenShift which can be helpful in validation). It also runs on IBM Power and mainframe if needed, showcasing versatility. VMware Tanzu ties into vSphere nicely. These platforms come with monitoring, logging, and service mesh out-of-box which is useful.

Data Management and Analytics Tools: Pharma datacenters use a variety of databases and analytics platforms: relational databases (Oracle, SQL Server, PostgreSQL) for transactional data; NoSQL databases (MongoDB, Cassandra) for certain unstructured data or caching; Hadoop/Spark clusters for big data processing (though a lot of that is moving to cloud managed services like Databricks or EMR). There are also domain-specific platforms: e.g., Lab data systems might use scientific data management systems (like PerkinElmer or Thermo Fisher platforms) that require backend servers and storage. Ensuring the datacenter can host these with required performance (often using high-memory servers for in-memory databases or GPU for analytics) is part of vendor planning. Tools like SAS are still heavily used in clinical analytics and pharmacovigilance, so SAS grids might be deployed on clusters – a modern trend is containerizing SAS or moving it to cloud, but many keep it on VMs for control.

Security Vendors: In addition to native cloud and open-source tools, pharma datacenters utilize products from security vendors: Palo Alto Networks or Check Point firewalls, CyberArk for privileged access, Splunk or IBM QRadar for SIEM, etc. As datacenters modernize, some are adopting **zero trust platforms** like Zscaler for secure access and microsegmentation tools like Illumio or Guardicore to map and enforce traffic flows between servers. The integration of these tools requires a flexible datacenter network and compute environment.



building AI capability often include NVIDIA DGX systems (essentially GPU supercomputers) or servers with multiple NVIDIA GPUs. NVIDIA's networking gear (InfiniBand switches and cards) is prevalent in HPC clusters. They also produce DPUs (data processing units – SmartNICs) which offload security and network tasks in the datacenter. NVIDIA's **BioNeMo platform**, significantly expanded in January 2026, is becoming the foundation for AI-driven drug discovery. The platform includes NVIDIA Clara open models such as RNAPro for RNA structure prediction and ReaSyn v2 for drug synthesis planning, along with BioNeMo Recipes for scaling biological foundation model training and nvMolKit for GPU-accelerated cheminformatics ^[41]. Major pharma partnerships include the \$1 billion NVIDIA-Eli Lilly co-innovation lab, GSK's London AI hub with DGX A100 systems, and Bristol-Myers Squibb's DGX SuperPOD deployment (which reportedly achieved 55% overall cost savings compared to prior computing infrastructure). In December 2025, Merck & Co. produced the KERMT model for small-molecule drug discovery using NVIDIA infrastructure. This combination of hardware and software makes NVIDIA a dominant player in the pharma computing ecosystem.

Another emerging vendor category is **cloud HPC/quantum startups** offering HPC-as-a-service or quantum-as-a-service. Global quantum technology industry funding surged to \$1.25 billion in Q1 2025 (a 125% increase), supporting faster molecular simulations, precision drug design, and AI-accelerated quantum workflows. Notably, Google opened its Sycamore v2 hardware to pharma users via cloud queues, enabling high-fidelity gate-based simulations without on-prem infrastructure ^[521]. Pharma IT increasingly monitors these for drug discovery applications like quantum chemistry simulations and optimization problems.

Comparison and Considerations:

- For **compute and storage**, Dell and HPE (with possibly Lenovo, IBM) are the main on-prem choices. Both have proven track records in pharma. The choice may come down to existing vendor relationships, specific performance needs (one might have an edge in a benchmark), or ecosystem (e.g., if a company likes VMware-based HCI, Dell VxRail (VMware+Dell) vs HPE vSAN Ready Nodes are both options).
- For **cloud**, AWS has slightly more pharma references historically, but Azure is extremely close, especially in companies already Microsoft-heavy. GCP is often used for specific projects (AI or data crunching) rather than whole IT outsourcing, but some smaller biotechs might go GCP-first because of cost or expertise availability. Multi-cloud strategy means many large pharmas use 2 or all 3 in different capacities.
- For **network**, Cisco vs Arista vs Juniper etc., often Cisco remains due to enterprise support and familiarity. But for new high-speed deployments, Arista's performance can be attractive.



with strong internal talent might use upstream Kubernetes or OpenStack.

Ultimately, a modern pharma datacenter is an **integration of multiple vendor solutions**: e.g., Dell servers running VMware and Red Hat OpenShift, connected by Cisco switches, storing data on NetApp or Dell EMC arrays, backing up to a Cohesity appliance, all monitored by Splunk, etc., and hybrid-connected to Azure and AWS for overflow. Interoperability and support agreements are crucial. Many vendors form partnerships (e.g., Dell and NVIDIA, or Microsoft and SAS) that ease integration. Pharma IT architects often evaluate total solution stacks rather than individual pieces. Vendor selection also involves considering future roadmaps (is the vendor investing in new tech? do they understand pharma's compliance needs?).

The ecosystem is rich, and **pharma-specific consulting firms** or system integrators (like Deloitte, Accenture, etc.) often assist in selecting and deploying the right mix of tools, which brings in another layer of vendor relationship.

In summary, no single vendor provides everything – it's about choosing the right mix. Cloud providers offer the agility and services; traditional IT vendors ensure on-prem remains robust and integrated; specialized vendors fill gaps (security, niche software). The **best practices** section next will touch on how to manage this mix during modernization.

Best Practices for Modernization and Digital Transformation in Pharma Datacenters

Modernizing a pharmaceutical datacenter is a multi-faceted journey. It's not just about new tech, but aligning people, processes, and regulatory compliance with that tech. Here are some **best practices and strategies** for pharma IT teams undertaking datacenter modernization and digital transformation:

1. Develop a Clear Modernization Roadmap

Start with a thorough assessment of the current state: inventory the infrastructure, map the applications (and which are GxP or mission-critical), and identify pain points (e.g., capacity bottlenecks, high support costs, technology end-of-life). Then create a roadmap that aligns IT improvements with business goals (faster research cycles, improved data sharing, cost reduction,



Larger projects could include building a new HPC cluster or implementing a global data lake. Having executive buy-in on this roadmap is crucial, especially since pharma CIOs must justify changes against regulatory risk concerns. A phased approach is often recommended: e.g., phase 1 – consolidate and virtualize, phase 2 – introduce hybrid cloud for dev/test, phase 3 – migrate GxP apps as confidence grows, etc.

2. Embrace Hybrid Cloud Thoughtfully

It's generally not practical or wise to try moving *everything* to cloud at once in a regulated environment. Instead, identify suitable candidates for cloud (such as data analysis workloads, new applications that aren't entangled with legacy systems, or external-facing apps). Implement those on cloud with proper security and validation. Simultaneously, refresh on-prem infrastructure for workloads staying in-house – possibly adopting technologies that make integration with cloud easier (like container platforms that work in both places). **Maintain consistent environments** where possible: using containerization and infrastructure-as-code so that whether a workload runs on-prem or on cloud, it follows the same deployment process. This consistency eases validation and troubleshooting. Many pharma companies set up a "**cloud center of excellence**" – a cross-functional team to develop cloud governance (security, cost management, compliance guidelines) and assist other teams in using cloud properly. That team ensures that as modernization proceeds, it's done in a compliant and secure way (for instance, verifying that any cloud usage involving health data has encryption and BAA in place, that data is retained appropriately, etc.). Also, plan for **data integration** in a hybrid model: use robust integration middleware or APIs so that on-prem and cloud systems can exchange data seamlessly (e.g., a cloud analytics platform pulling data from an on-prem Oracle database via a secure API).

3. Ensure Compliance and Validation from Day One

When modernizing, loop in the Quality Assurance/Compliance specialists early. If you're adopting a new platform (say Kubernetes or a new SDN tool), perform a **vendor audit** if necessary and understand how to validate it. Follow frameworks like GAMP 5 (Good Automated Manufacturing Practice) which provides guidelines for validating computerized systems in pharma. GAMP 5 encourages a risk-based approach – focus validation efforts on critical aspects that could affect product quality or patient safety. For instance, infrastructure components that host GxP applications should be qualified (IQ/OQ/PQ). You can leverage vendor documentation and testing to support this. Many modern tools allow exporting configuration and test results – for example, if using Terraform scripts to deploy infrastructure, those scripts and their execution logs can serve as



pharma IT teams are adopting DevOps but with "DevSecOps" or "Dev(Q)Ops" mindset – embedding quality checks in the CI/CD pipeline. MasterControl (a compliance software firm) suggests that DevOps principles can be applied in life sciences by integrating compliance checks into the development pipeline ([53]). Documentation is key – update your SOPs (Standard Operating Procedures) to cover new processes (e.g., how to manage cloud changes under change control). Train your IT staff and even end-users on any new responsibilities (for instance, if scientists will use a self-service portal to spin up cloud VMs, they should understand the rules and shared responsibility).

4. Strengthen Security Posture in Tandem

Every modernization step should include a security evaluation. For example, if introducing container platforms, ensure image scanning and cluster security policies are in place. If extending network to cloud, implement strong encryption (VPNs or private links) and consider a zero-trust network approach (no implicit trust for on-prem vs cloud, authenticate and authorize each transaction). Use modern identity solutions – possibly moving to a unified cloud-friendly identity provider (like Azure AD) so that MFA and conditional access policies can be enforced uniformly for any app. Also consider **segmentation**: as you virtualize or containerize, it's an opportunity to segment networks at finer granularity (using NSX or cloud security groups). This can actually improve security compared to legacy flat networks. Employing **automation in security** is a best practice – e.g., Infrastructure as Code means you can incorporate security checks as code. If a developer tries to deploy a storage bucket without encryption, an automated check can flag or prevent it. Some pharma companies invest in automated compliance tools that continually scan configurations against policies (for example, ensuring no open ports that shouldn't be, all systems are patched, etc.). Given the stat that human error causes most breaches ([54]), automation and *fail-safe defaults* (secure by default configurations) are vital.

5. Leverage High-Performance Technologies to Future-proof

Modernization in pharma often involves upgrading to newer technologies that can handle future demands – e.g., adopting NVMe storage, 100 Gbit networks, GPU acceleration, etc. It's wise to **build headroom** into the design. If doing a tech refresh of servers, choosing ones that have extra RAM slots or support next-gen processors can extend their life. Similarly, when building out the network, maybe deploy fiber that can support higher speeds later or modular switches that can be upgraded. Considering **modularity and scalability** as design criteria ensures the datacenter can grow without a complete rehaul each time. Hyper-converged systems are easy to scale by adding



just keep adding commodity nodes. Also keep an eye on **emerging tech**: maybe not adopt immediately, but be prepared. For example, **AI Ops tools** (AIOps) that use AI to manage infrastructure can help in future to automate troubleshooting and optimize resource allocation – adopting monitoring systems now that have that future capability would be beneficial. Likewise, **serverless computing** and event-driven architectures might play a bigger role in pharma IT (for things like data processing pipelines); experimenting in non-critical areas now can prepare the organization. Essentially, avoid painting yourself into a corner with proprietary or inflexible tech. Favor open standards and interoperability so you can plug in new components as needed.

6. Enhance Data Architecture and Enable Advanced Analytics

Modernization is a chance to break down silos. Implement a robust **data architecture** where data from R&D, clinical, manufacturing can come together (with proper governance) for holistic analysis. Many pharma companies are building **data lakes** or analytics platforms as part of digital transformation. Best practice is to classify data (identify which data is sensitive/personal) and handle accordingly (tokenize or restrict access to raw PHI, etc., while still making aggregated data available for analysis). Use modern ETL/ELT tools to ingest data from legacy systems into the lake. Then provide self-service analytics tools or sandboxes for data scientists – possibly via a cloud-based analytics workspace. For example, setting up a secure enclave in the cloud with de-identified data where data scientists can apply AI algorithms without risking primary data. This way, modernization directly empowers AI/ML initiatives, which are often at the heart of digital transformation. A concrete best practice is deploying a platform for reproducible research computing – e.g., JupyterHub or RStudio Server on the Kubernetes cluster, integrated with version control – to enable scientists to use datacenter resources in a controlled yet flexible manner. This ties IT modernization to the productivity of scientists and speeds up innovation (since IT is no longer a bottleneck but an enabler).

7. Foster Cross-functional Collaboration and Skills Development

Technology change must be accompanied by people change. Encourage collaboration between IT, R&D, and compliance teams. For instance, form a **cloud steering committee** with representatives from security, compliance, R&D computing, etc., to guide cloud adoption so everyone's requirements are heard. Invest in **training**: upskill IT staff on new technologies (container orchestration, cloud architecture, IaC tools, etc.), and conversely educate scientists and business users on how to use new IT services (like a new data portal or analytic tool). Embrace DevOps culture where appropriate – breaking silos between development and operations. In pharma, this



brings a software engineering approach to managing infrastructure, focusing on automation and reliability metrics. All these cultural shifts require management support and possibly new KPIs (e.g., measuring deployment frequency, mean time to recover, etc., instead of just traditional uptime, to gauge improvement through DevOps).

8. Incremental Validation and Pilot Programs

When introducing a significant change, pilot it in a low-risk area first. For example, before rolling out containerization for GxP apps, pilot Kubernetes with a non-GxP app or in a test environment to iron out issues and build SOPs. When adopting hybrid cloud, maybe start with development and test environments in cloud while production remains on-prem, until confidence and experience are gained (this also yields performance and cost data to adjust plans). Use **agile methodology** for IT projects: iterate in sprints, demonstrate results to stakeholders frequently, and adjust the plan based on feedback. This is somewhat new in infrastructure projects but is being adopted – e.g., instead of an 18-month project to overhaul everything, deliver incremental improvements every few months (such as migrating one department's workloads to new platform, then the next, etc.). Each increment should conclude with a review of compliance and performance to ensure it meets requirements before proceeding.

9. Monitor, Measure, Optimize

Once new systems are in place, continuously monitor both technical metrics and process metrics. For technical: use monitoring dashboards for infrastructure health, capacity, and also for new metrics like cloud spending (to ensure cost optimization). For process: track how long it takes to provision a new environment now versus before, or how often releases happen for an app – indicators of improved agility. Collect user feedback – are scientists finding the new HPC cluster faster? Did a new collaboration tool actually help or is it cumbersome? Use this data to optimize configurations, provide additional training, or even backtrack on decisions that didn't pan out. Modernization is not a one-time event; it's ongoing. By creating a feedback loop, the datacenter continuously evolves. Many organizations use **maturity models** (there's Pharma 4.0 maturity model aligned with Industry 4.0) to assess where they stand and identify next steps in digital maturity. Benchmark against peers if possible – forums and consortia in pharma IT (like BioPhorum, Pistoia Alliance) can provide insight on what others are doing and what's working.

10. Focus on Resilience and Continuity



assume any component can fail and plan redundancy (multi-AZ deployment in cloud, active-active datacenters on-prem, etc.). Automate backups and test restores regularly (especially important when adopting new tech – ensure you can get data out of a cloud service or that your container storage is backed up, etc.). Disaster Recovery plans should be updated to reflect the new environment – for instance, if workloads moved to cloud, your DR strategy might shift to failing over to a different region or rolling back to on-prem if cloud has an outage. Regular drills and tabletop exercises involving IT and business stakeholders keep everyone prepared. This is a best practice because transformation should not come at the cost of reliability – regulators and patients expect high availability of critical systems (pharmacovigilance databases, for example, must be up so adverse events are processed in a timely manner). Use the modernization effort to also modernize DR – maybe move from tape backups to cloud backups with quicker restore, etc.

By following these best practices, pharmaceutical companies can modernize their datacenters in a **controlled, compliant, and effective** way. The end result should be an IT environment that is far more agile and scalable, able to support advanced research and analytics, while still maintaining the rigor of compliance and the robustness of security that the pharma industry demands. It's truly a journey – modernization is continuous – but each step brings new capabilities that can help bring therapies to patients faster and improve operational efficiency in this critical industry.

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